

HAT-P-12b

R-1355

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_210411 · created 2026-07-31T23:01:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459315.9024 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.032
Transit depth (Rp/Rs)^2	2.06 +/- 0.92 [%]
Orbital Inclination (inc)	85.83 +/- 1.32
Airmass coefficient 1 (a1)	0.924 +/- 0.012
Airmass coefficient 2 (a2)	0.069 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	#3 - [430, 144]
Optimal Method	PSF photometry
Transit Duration (day)	0.058 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.144 ± 0.032 vs 0.1406 (deviation 0.11σ)
Duration fit vs published	0.058 d vs 0.0975 d (deviation 1.32σ)
Mid-time O–C	8.6 min
Depth SNR	4.5
Residual scatter	1.35 %

Light curve

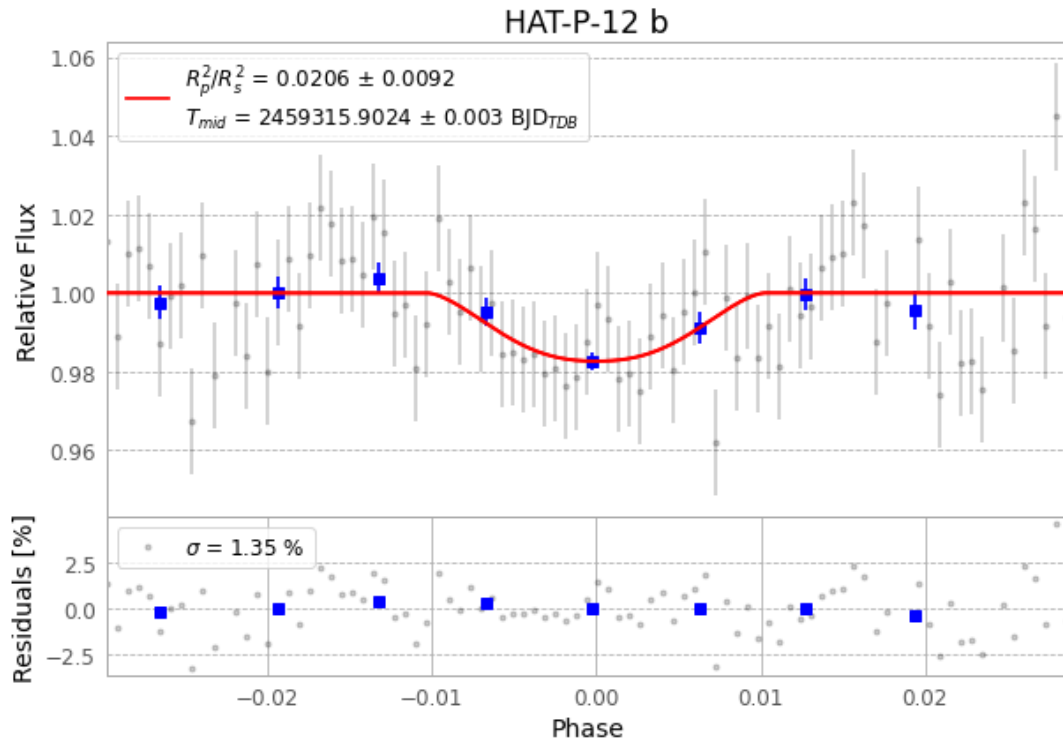


Figure 1 — FinalLightCurve_HAT-P-12 b_2021-04-11.png (SHA-256 e0780c0928be74b5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [407, 215], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6c32086ea6b4fce6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), HATP-12210411071514.FITS → HATP-12210411114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-210411.log (3ca1c541255b...), FinalLightCurve_HAT-P-12 b_2021-04-11.png (e0780c0928be...), FinalLightCurve_HAT-P-12 b_2021-04-11.csv (ce351ab08f1b...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 23:01Z.